Teaching notes and solutions

> **Teacher note · Teacher notes.** Use two different parameters to compare two lines. Check all three coordinates, even if two suffice to calculate candidates. A Cartesian equation on its own generally describes a plane in space; it does not describe a line. Reuse the projection from Chapter 11 to give meaning to the solving of systems.

Solutions to the activity

1. A point of d_1 has coordinates (t, t, t). It belongs to the plane if and only if $3t = 3$, that is $t = 1$. The intersection is therefore the single point $(1, 1, 1)$.
2. For a point of d_2, $x + y + z = 3 + t - t + 0 = 3$ for every t. All points of the line are in the plane: $d_2 \subset \mathcal{P}$.
3. For a point of d_3, $x + y + z = t - t + 0 = 0$. The equation $0 = 3$ has no solution: the line is disjoint from the plane. Its direction vector $(1, -1, 0)$ is orthogonal to the normal $(1, 1, 1)$; it is parallel to the plane.
4. After substitution, we obtain respectively an equation with a single solution, an identity true for every parameter value, and a contradiction. These three cases correspond to a single intersection point, containment, and no intersection.

E1. $\overrightarrow{AB} = (2, 1, -1) \neq \vec{0}$. The line therefore has the representation $(x, y, z) = (1, -1, 2) + t(2, 1, -1)$, $t \in \mathbb{R}$. The values $t = 0$ and $t = 1$ give back A and B.

E2. For $A(1, 2, 3)$, the condition $\overrightarrow{AM} \cdot (1, 0, -2) = 0$ is $(x - 1) - 2(z - 3) = 0$. After expanding, the plane has equation $x - 2z + 5 = 0$. The point A satisfies it.

E3. After substitution, $t + 2(1 + t) + (2 - t) = 8$, that is $2t + 4 = 8$, so $t = 2$. The unique intersection is $(2, 3, 0)$; we check $2 + 2 \times 3 + 0 = 8$.

E4. The second equation gives $x = y$. Let $x = y = t$; the first becomes $2t + z = 1$. The set of solutions is $(x, y, z) = (t, t, 1 - 2t) = (0, 0, 1) + t(1, 1, -2)$, a line.

E5. The normals are collinear, the second being twice the first. But doubling $2x - y + z = 3$ gives $4x - 2y + 2z = 6$, which is incompatible with the right-hand side 5. The planes are therefore strictly parallel, not intersecting.

Solutions to Homework 12

1. The coefficients of x, y, z give the non-zero normal $\vec{n} = (1, 2, 2)$ in the orthonormal frame.
2. Look for H on the normal line through M: $H = (3 + t, 2t, 2t)$. Membership of $\mathcal{P}$ gives $3 + t + 4t + 4t = 9$, so $t = \frac{2}{3}$. Thus $H = (\frac{11}{3}, \frac{4}{3}, \frac{4}{3})$. We check $\frac{11}{3} + \frac{8}{3} + \frac{8}{3} = 9$ and $\overrightarrow{MH} = (\frac{2}{3})\vec{n}$. So H is indeed the orthogonal projection, and $MH = (\frac{2}{3})\sqrt{1 + 4 + 4} = 2$.
3. The normals $(1, 2, 2)$ and $(1, -1, 0)$ are not collinear: the planes meet along a line. Since $x = y$, let $x = y = s$. The first equation gives $3s + 2z = 9$, that is $z = \frac{9-3s}{2}$. The intersection is the line $(x, y, z) = (s, s, \frac{9-3s}{2})$, $s \in \mathbb{R}$.
4. The point H has $x_H = \frac{11}{3} \neq \frac{4}{3} = y_H$, so it belongs neither to $\mathcal{Q}$ nor to the line of intersection.
5. The cote is the z-coordinate. Setting it to zero requires $9 - 3s = 0$, so $s = 3$ and $K = (3, 3, 0)$. Then $MK = 3 > 2 = MH$. This result is consistent with the property of the orthogonal projection: among all points of the plane, H is the closest to M.

Solutions to the textbook or past-paper work

1. True: the line passes through B and its direction vector $(-2, -1, 3)$ equals $-\frac{1}{2}\overrightarrow{AB}$.
2. False: $\vec{n} \cdot \overrightarrow{OB} = 15 - 4 - 1 = 10 \neq 0$.
3. False: for $k = -4$ and $s = \frac{5}{2}$, the two representations give $(11, 12, -14)$. The lines intersect, so they are coplanar.
4. True: $d(C, \mathcal{P}) = \frac{|2-(-1)+2+1|}{\sqrt{1+1+1}} = 2\sqrt{3}$.